CSE351 Computer Networks

Programming Assignment #4 Report

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1. Code Implementation
   a) sr_router.c
      1) validation & firewall
         Validates the packet by checking its length and IP checksum. And also Blocks packets related to the blacklisted subnet (10.0.2.0/24) via the ip_black_list function.
      2) classification & processing
         Destined for the Router itself:
         Responds with an Echo Reply if an ICMP Echo Request (Ping) is received. Sends an ICMP Port Unreachable (Type 3, Code 3) message if a TCP/UDP packet is received, as the ports are closed.

         Destined for External Network (Forwarding):

         Checks the TTL; if it is 1 or less, sends an ICMP Time Exceeded (Type 11) message and drops the packet. Performs a Longest Prefix Match (LPM) on the routing table. If no route is found, sends an ICMP Net Unreachable (Type 3, Code 0) message.

      3) Forwarding
         Decrements the TTL and recalculates the checksum. Checks the ARP cache for the Next Hop's MAC address. If found, sends the packet immediately; otherwise, queues the packet and schedules an ARP request.

   b) sr_arpcache.c
      1) retransmission
         Checks ARP requests approximately every second. If there is a request that has not yet received a reply, it re-broadcasts the ARP Request packet.
      2) Failure Handling
         If an ARP request receives no reply after 5 attempts, it determines that the host is unreachable. Sends an ICMP Destination Host Unreachable (Type 3, Code 1)

message to the sender for all packets waiting in the queue for that ARP request, and then destroys the request.

2. Execution Results

a) ARP success case

Figure1 shows that the implementation successfully performed ARP resolution, and the ICMP Echo Request was processed correctly.

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	02:d3:9b:b3:2b:7a	Broadcast	ARP	42	Who has 10.0.1.1? Tell 10.0.1.100
2	0.000169	3e:d9:8d:86:d3:b7	02:d3:9b:b3:2b:7a	ARP	42	10.0.1.1 is at 3e:d9:8d:86:d3:b7
3	0.006102	10.0.1.100	192.168.2.2	ICMP	98	Echo (ping) request id=0x661b, seq=1/256, ttl=64 (no response found!)
4	0.006136	3e:39:3b:a0:98:c3	Broadcast	ARP	42	Who has 192.168.2.2? Tell 192.168.2.1
5	0.008466	ca:4d:2c:3e:e3:79	3e:39:3b:a0:98:c3	ARP	42	192.168.2.2 is at ca:4d:2c:3e:e3:79
6	0.008569	10.0.1.100	192.168.2.2	ICMP	98	Echo (ping) request id=0x661b, seq=1/256, ttl=63 (reply in 7)
7	0.011753	192.168.2.2	10.0.1.100	ICMP	98	Echo (ping) reply id=0x661b, seq=1/256, ttl=64 (request in 6)
8	0.011769	3e:d9:8d:86:d3:b7	Broadcast	ARP	42	Who has 10.0.1.100? Tell 10.0.1.1
9	0.014187	02:d3:9b:b3:2b:7a	3e:d9:8d:86:d3:b7	ARP	42	10.0.1.100 is at 02:d3:9b:b3:2b:7a
10	0.014222	192.168.2.2	10.0.1.100	ICMP	98	Echo (ping) reply id=0x661b, seq=1/256, ttl=63
11	1.570552	10.0.1.100	192.168.2.2	ICMP	98	Echo (ping) request id=0x6633, seq=1/256, ttl=64 (no response found!)
12	1.570586	10.0.1.100	192.168.2.2	ICMP	98	Echo (ping) request id=0x6633, seq=1/256, ttl=63 (reply in 13)
13	1.579967	192.168.2.2	10.0.1.100	ICMP	98	Echo (ping) reply id=0x6633, seq=1/256, ttl=64 (request in 12)
14	1.580301	192.168.2.2	10.0.1.100	ICMP	98	Echo (ping) reply id=0x6633, seq=1/256, ttl=63
15	5.101939	ca:4d:2c:3e:e3:79	3e:39:3b:a0:98:c3	ARP	42	Who has 192.168.2.1? Tell 192.168.2.2
16	5.102196	3e:39:3b:a0:98:c3	ca:4d:2c:3e:e3:79	ARP	42	192.168.2.1 is at 3e:39:3b:a0:98:c3

Figure 1.

b) Other ICMP

Figure 2 demonstrates that the router correctly handles the ARP failure scenario. After five unsuccessful ARP requests, the router generates an ICMP Destination Host Unreachable (Type 3, Code 1) message as required.

1	0.000000	10.0.1.100	192.168.2.50	ICMP	98	Echo (ping) request id=0x68cb, seq=1/256, ttl=64 (no response found!)
2	0.000157	3e:39:3b:a0:98:c3	Broadcast	ARP	42	Who has 192.168.2.50? Tell 192.168.2.1
3	0.956359	3e:39:3b:a0:98:c3	Broadcast	ARP	42	Who has 192.168.2.50? Tell 192.168.2.1
4	1.958737	3e:39:3b:a0:98:c3	Broadcast	ARP	42	Who has 192.168.2.50? Tell 192.168.2.1
5	2.977435	3e:39:3b:a0:98:c3	Broadcast	ARP	42	Who has 192.168.2.50? Tell 192.168.2.1
6	3.993785	3e:39:3b:a0:98:c3	Broadcast	ARP	42	Who has 192.168.2.50? Tell 192.168.2.1
7	5.201014	02:d3:9b:b3:2b:7a	3e:d9:8d:86:d3:b7	ARP	42	Who has 10.0.1.1? Tell 10.0.1.100
8	5.202319	3e:d9:8d:86:d3:b7	02:d3:9b:b3:2b:7a	ARP	42	10.0.1.1 is at 3e:d9:8d:86:d3:b7
9	6.029166	3e:d9:8d:86:d3:b7	Broadcast	ARP	42	Who has 10.0.1.100? Tell 10.0.1.1
10	6.058312	02:d3:9b:b3:2b:7a	3e:d9:8d:86:d3:b7	ARP	42	10.0.1.100 is at 02:d3:9b:b3:2b:7a
11	6.066974	192.168.2.1	10.0.1.100	ICMP	70	Destination unreachable (Host unreachable)

Figure 2

Figure3 shows the router successfully received the UDP packet and correctly responded with an ICMP Port Unreachable message.

17	2.133808	10.0.1.100	10.0.1.1	UDP	47	53877 → 9999 Len=5
18	2.134189	10.0.1.1	10.0.1.100	ICMP	70	Destination unreachable (Port unreachable)
19	2.152595	10.0.1.100	10.0.1.1	UDP	74	47789 → 33434 Len=32
20	2.152715	10.0.1.1	10.0.1.100	ICMP	70	Destination unreachable (Port unreachable)
21	2.152888	10.0.1.100	10.0.1.1	UDP	74	33226 → 33435 Len=32
22	2.152897	10.0.1.1	10.0.1.100	ICMP	70	Destination unreachable (Port unreachable)
23	2.160168	10.0.1.100	10.0.1.1	UDP	74	37425 → 33436 Len=32
24	2.160190	10.0.1.1	10.0.1.100	ICMP	70	Destination unreachable (Port unreachable)
25	2.171960	10.0.1.100	4.4.4.4	ICMP	98	Echo (ping) request id=0x6a57, seq=1/256, ttl=64 (no response found!)
26	2.172125	10.0.1.1	10.0.1.100	ICMP	70	Destination unreachable (Network unreachable)

Figure 3

Using ping binary with -t 1 option, I checked the router correctly handled packets with expired TTL values by generating an ICMP Time Exceeded message. Check Figure4.

1	0.000000	10.0.1.100	192.168.2.2	ICMP	98	Echo (ping) request id=0x6c1b, seq=1/256, ttl=1 (no response found!
2	0.000055	3e:d9:8d:86:d3:b7	Broadcast	ARP	42	Who has 10.0.1.100? Tell 10.0.1.1
3	0.005260	02:d3:9b:b3:2b:7a	3e:d9:8d:86:d3:b7	ARP	42	10.0.1.100 is at 02:d3:9b:b3:2b:7a
4	0.005308	10.0.1.1	10.0.1.100	ICMP	70	Time-to-live exceeded (Time to live exceeded in transit)

Figure 4

Figure 5 demonstrates that the router correctly handles packets destined for unknown networks. When no matching route is found, the router generates an ICMP Destination Unreachable (Type 3, Code 0) message.

1	0.000000	02:d3:9b:b3:2b:7a	Broadcast	ARP	42	Who has 10.0.1.1? Tell 10.0.1.100
2	0.000076	3e:d9:8d:86:d3:b7	02:d3:9b:b3:2b:7a	ARP	42	10.0.1.1 is at 3e:d9:8d:86:d3:b7
3	0.010243	10.0.1.100	8.8.8.8	ICMP	98	Echo (ping) request id=0x6ecd, seq=1/256, ttl=64 (no response found!)
4	0.010273	3e:d9:8d:86:d3:b7	Broadcast	ARP	42	Who has 10.0.1.100? Tell 10.0.1.1
5	0.060925	02:d3:9b:b3:2b:7a	3e:d9:8d:86:d3:b7	ARP	42	10.0.1.100 is at 02:d3:9b:b3:2b:7a
6	0.060977	10.0.1.1	10.0.1.100	ICMP	70	Destination unreachable (Network unreachable)

Figure 5

c) Firewall

My firewall successfully blocked.

```
mininet> client2 ping -c 1 192.168.2.2
PING 192.168.2.2 (192.168.2.2) 56(84) bytes of data.

--- 192.168.2.2 ping statistics ---
1 packets transmitted, 0 received, 100% packet loss, time 0ms

mininet>
```

```
eth4    MAC Addr: 0a:28:17:f3:b1:f7
         IP Addr: 10.0.2.1
eth3    MAC Addr: 3e:d9:8d:86:d3:b7
         IP Addr: 10.0.1.1
eth2    MAC Addr: b2:94:2d:e9:9f:4d
         IP Addr: 172.64.3.1
eth1    MAC Addr: 3e:39:3b:a0:98:c3
         IP Addr: 192.168.2.1
 <-- Ready to process packets -->
[IP blocked] : 10.0.2.100
```

Reference

D. Plummer, "An Ethernet Address Resolution Protocol: Or Converting Network Protocol Addresses to 48.bit Ethernet Address for Transmission on Ethernet Hardware," RFC 826, Nov. 1982.

J. Postel, "Internet Protocol," RFC 791, Sep. 1981.

J. Postel, "Internet Control Message Protocol," RFC 792, Sep. 1981.